

Dr. Onur Çatmabacak

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PROFESSIONAL SUMMARY

Detail-oriented and results-driven **software developer** with 3 years of industry experience in product development. I have been a **Linux** enthusiast since 2008. Proficient in **bash** and **python** programming languages. Throughout my career, I have successfully developed and deployed commercial software that finds the minimum 2D surfaces to support a 3D object for additive manufacturing.

I am experienced in **agile methodologies**, having worked in fast-paced environments where I consistently delivered projects on time and within budget. I am skilled at translating business requirements into technical specifications and possess strong problem-solving abilities to overcome complex development challenges. Furthermore, I am committed to writing clean, maintainable, and well-documented code, following best practices and industry standards. I continuously stay updated with emerging technologies and tools to enhance my development skills and stay at the forefront of innovation.

As a software developer, I am driven by a constant desire to learn, grow, and take on new challenges. I am confident that my technical expertise, combined with my problem-solving skills and dedication, make me a valuable asset to your team.

EXPERIENCE

System Engineer

Nov. 2023 - Present

Bank of Julius Baer

Zurich, CH

- Collaborate with a team of four to integrate data from various applications across Bank Julius Baer into Splunk, enhancing data management and analysis capabilities.
- Responsible for developing, maintaining, and enhancing dashboards, reports, and alerts within Splunk to improve data visibility and facilitate informed decision-making processes.
- Utilize the TrackMe add-on to monitor the health and stability of the bank's servers and infrastructure, ensuring optimal performance and reliability.
- Provide specialized technical support to various departments within the bank, demonstrating strong teamwork, a service-oriented approach, and effective communication skills across interdisciplinary teams.

Software Engineer

Nov. 2021 – Jan. 2023

Spherene AG

Zurich, CH

- Replaced pymesh library with trimesh and many other various mesh libraries since the software license is not suitable for closed source applications and updated the python version of the whole application from 3.8.8 to 3.10.7. As a result, we did not pay +50k CHF for software licenses.
- Developed features, e.g. http1 to http2 upgrade and , for the back-end of our online service for additive manufacturing.
- Created a jupyter notebook utilizing the proprietary software developed by Spherene AG, running in a docker container on Google Cloud for an ESA project.
- Created projects running our closed-source additive manufacturing software on Google Cloud for the web service.
- Integrated FEniCS FEM Solver instead of NGSolve into our proprietary software.
- Replaced many packages with inconvenient software licenses for closed-source applications.
- Improved numerical algorithm of our closed-source additive manufacturing software.

PhD Researcher

Sep. 2016 – Dec. 2020

Institute for Computational Science, University of Zurich

Zurich, CH

- Developed a pipeline using Python to analyze large datasets of state-of-the-art cosmological simulations of the Universe from FIRE (Feedback In Realistic Environments) project that were run in various supercomputing centers (CSCS, XSEDE, BSC-CNS, Flatiron Institute, etc...) around the World.
- Proposed a theoretical model to explain the black hole - galaxy mass scaling relation.

- Created galaxy catalogs using Amiga Halo Finder (AHF), which is used by many other researchers around the world.
- Supervised our research group in IT, software development in Python/C/C++, and data analysis.
- The list of scientific papers produced in this Ph.D. can be found **IN THIS LINK**.

Supervised Projects

Institute for Computational Science, University of Zurich

Zurich, CH

- Artem Basyrov, Theoretical modeling of halo accretion, Master of Science Thesis Project, Feb. 2019 – May 2019
- Eric Rohr, The galaxy–halo size relation at cosmic noon, Think Swiss Summer Exchange Student Project, June 2019 – Sep. 2019

Teaching Assistant

Sep. 2016 – Feb. 2020

Institute for Computational Sciences, University of Zurich

Zurich, CH

- Teaching assistant for Theoretical Astrophysics, The Universe, Computational Astrophysics, Introduction to Data Science, Introduction to Computer Simulations courses.
- Responsible for problem-solving sections and correcting/grading weekly home-works.

Teaching Assistant

Sep. 2014 – June 2016

Faculty of Engineering and Natural Sciences, Sabanci University

Istanbul, TR

- Teaching assistant for Nature of Science course with ~ 750 students.
- Responsible for solving sessions and grading exams.

EDUCATION

University of Zurich

Zurich, CH

Ph.D. in Computational Astrophysics

Sep. 2016 – Oct. 2022

Thesis Title: The growth of supermassive black holes and their host galaxies in cosmological simulations.

Kültür University

Istanbul, TR

M.Sc. in Physics, GPA:3.45

Sep. 2011 – June 2014

Thesis Title: Accretion disks around weakly magnetized neutron star low-mass x-ray binaries.

Kültür University

Istanbul, TR

B.Sc. in Physics with Honours, GPA:3.69

Sep. 2007 – June 2011

Thesis Title: Radial dependent one-dimensional analytical solution of viscous accretion disks around neutron star low-mass x-ray binaries.

SKILLS

Languages: Turkish (Native), English (C1), German (B1)

Coding: Python ●●●●●, Django ●●●●●, BASH ●●●●●, FORTRAN95 ●●●●●, C/C++ ●●●●●, Julia ●●●●●

Dev Ops: Google Cloud ●●●●●, Docker ●●●●●, Git ●●●●●

Misc: Software License ●●●●● Linux ●●●●●

(**Grading:** Beginner ●●●●● Familiar ●●●●● Intermediate ●●●●● Comfortable ●●●●● Expert ●●●●●)